

WINDHAM, CT, USA

WMO: 725084

Lat: 41.742N

Lon: 72.184W

Elev: 247

StdP: 14.57

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54767

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.1	(c) 8.9	(d) -10.1	(e) 3.2	(f) 8.6	(g) -5.9	(h) 4.1	(i) 12.8	(j) 22.9	(k) 32.9	(l) 20.3	(m) 29.8	(n) 4.5	(o) 290	(p) 0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.8	73.4	87.1	72.1	84.2	70.4	76.4	84.9	75.0	82.4	73.4	80.0	8.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	126.9	79.9	72.6	121.8	78.5	71.7	118.1	77.6	40.0	85.0	38.5	82.7	37.2	80.2	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	17.0	15.2	-3.5	94.9	5.3	2.8	-7.3	96.9	-10.4	98.5	-13.4	100.1	-17.2	102.1
			-4.4	79.5	5.1	2.2	-8.0	81.0	-11.0	82.3	-13.8	83.5	-17.5	85.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	27.6	30.5	37.5	48.7	59.0	67.2	72.9	71.2	64.3	52.7	42.3	33.2
	DBStd	17.54	10.50	9.22	9.00	7.88	7.33	6.46	5.08	5.35	6.74	8.15	8.34	9.23
	HDD50	2616	696	549	400	113	8	0	0	0	66	258	526	
	HDD65	5903	1160	967	852	492	216	51	4	9	94	391	681	986
	CDD50	2870	1	2	13	75	288	516	709	656	429	149	27	5
	CDD65	682	0	0	0	4	31	117	248	201	73	8	0	0
	CDH74	6017	0	0	8	81	417	1096	2166	1656	543	49	1	0
	CDH80	1806	0	0	1	26	129	342	712	473	119	4	0	0
	WSAvg	5.6	6.0	6.3	7.1	6.8	6.1	5.1	4.7	4.2	4.6	5.3	5.5	5.5
	PrecAvg	48.0	3.7	3.3	4.2	4.1	3.6	4.1	3.5	4.0	4.4	4.1	3.9	4.6
Precipitation	PrecMax	59.4	7.8	7.6	10.7	9.4	7.3	11.1	10.6	10.4	9.5	14.1	7.1	8.5
	PrecMin	32.8	1.8	1.5	0.5	1.2	1.0	0.2	1.5	1.6	1.0	0.6	0.9	1.6
	PrecStd	6.9	1.5	1.3	2.4	2.1	1.4	2.9	1.7	2.2	2.2	3.0	1.5	1.6
	PrecStd	6.9	1.5	1.3	2.4	2.1	1.4	2.9	1.7	2.2	2.2	3.0	1.5	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	60.0	61.3	73.3	84.2	89.8	91.2	93.5	91.3	88.3	79.3	69.7	63.9	
	0.4% MCWB	55.8	53.5	59.1	64.2	68.6	73.8	75.9	75.6	73.3	66.3	59.5	57.6	
	2% DB	54.0	54.8	63.5	73.9	83.4	87.8	89.9	87.9	83.1	74.3	64.4	57.7	
	2% MCWB	49.7	49.4	51.7	56.4	66.3	72.0	74.5	73.5	69.5	65.0	57.7	54.3	
	5% DB	47.6	49.6	56.9	68.4	78.9	83.9	87.0	84.8	80.2	70.7	61.3	53.2	
	5% MCWB	43.4	43.4	48.3	54.0	63.4	69.6	72.8	71.4	68.4	63.3	54.9	49.8	
	10% DB	42.7	44.8	52.3	63.5	73.4	80.9	84.0	82.2	76.7	66.6	57.2	47.8	
	10% MCWB	37.9	39.0	45.1	51.4	60.6	67.9	71.0	70.1	66.8	60.2	51.2	43.3	
	0.4% WB	56.9	56.6	59.9	65.2	72.0	76.7	78.8	78.6	75.6	71.3	63.8	59.9	
	0.4% MCDB	58.9	59.7	70.6	79.7	84.5	87.5	89.6	87.7	82.4	74.7	66.8	62.9	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	50.5	50.0	54.4	61.3	68.9	74.4	76.6	76.3	73.3	67.9	60.2	54.7	
	2% MCDB	52.8	53.6	59.9	70.0	78.7	83.8	86.0	83.1	78.7	72.1	63.5	57.6	
	5% WB	43.7	43.8	50.1	57.7	66.2	72.3	75.1	74.7	71.8	64.6	56.7	49.7	
	5% MCDB	47.0	48.5	55.9	64.4	75.3	80.3	83.0	80.9	76.9	68.8	60.3	52.5	
	10% WB	38.4	39.4	45.7	54.0	63.5	70.2	73.5	73.0	69.9	61.4	52.3	43.9	
	10% MCDB	41.5	43.9	51.3	60.6	69.9	76.9	80.2	79.0	74.6	66.1	56.4	47.3	
Mean Daily Temperature Range	MDBR	17.1	18.4	19.0	21.5	21.5	20.1	19.9	19.7	20.8	20.3	19.1	16.7	
	5% DB MCDBR	22.5	24.9	27.2	30.5	29.4	25.1	22.7	21.7	23.4	23.5	23.7	22.6	
	5% DB MCWBR	19.3	19.3	17.5	15.6	13.9	11.9	9.7	9.2	11.9	14.7	17.7	20.1	
	5% WB MCDBR	22.2	23.4	24.2	24.2	24.2	21.5	18.9	17.5	17.8	19.3	20.4	21.2	
	5% WB MCWBR	20.3	20.3	18.2	15.1	13.0	11.3	9.0	8.6	10.8	14.5	17.9	20.3	
	5% WB MCWBR	20.3	20.3	18.2	15.1	13.0	11.3	9.0	8.6	10.8	14.5	17.9	20.3	
Clear-Sky Solar Irradiance	taub	0.295	0.304	0.328	0.370	0.414	0.457	0.476	0.443	0.372	0.335	0.308	0.294	
	taud	2.479	2.465	2.462	2.373	2.279	2.195	2.150	2.231	2.408	2.526	2.559	2.554	
	Ebn at Noon	273	288	293	285	274	261	254	260	273	273	267	265	
	Edh at Noon	23	28	31	37	42	45	47	42	33	26	22	20	
All-Sky Solar Radiation	RadAvg	588	861	1176	1518	1686	1834	1888	1686	1384	920	640	486	
	RadStd	46	77	87	163	173	184	106	90	104	88	50	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	N/A
(47)	Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	+90

Nomenclature: See separate page